

LORENCE B. OJALES

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PROFESSIONAL SUMMARY

Computer Engineering graduate (Cum Laude) with hands-on experience in embedded systems, IoT development, robotics, and software engineering. Experienced in developing ESP32 and Raspberry Pi-based solutions involving firmware development, sensor integration, hardware-software integration, RESTful APIs, and database systems. Proficient in C/C++, Python, SQL, Git, Docker, and Linux environments. Strong problem-solving, debugging, and system integration skills developed through academic, hackathon, and independent engineering projects.

WORK EXPERIENCE

IT Technical Support Intern, JTEKT Philippines Corporation June 2025 – August 2025

- Provided technical support and troubleshooting for hardware, software, and network issues.
- Assisted in server monitoring, system maintenance, and IT infrastructure operations.
- Installed and configured computer systems while supporting end users and IT documentation.

PROJECTS

Eco Flow: AI-Powered Smart Greenhouse Management System (Thesis Project) 2025 - 2026

Technologies: Raspberry Pi 5, ESP32, Python (Flask), SQLite, Scikit-learn, Arduino IDE

- Developed an IoT-based greenhouse monitoring system for real-time sensor monitoring and control.
- Designed RESTful APIs using Flask for device communication and data management.
- Implemented machine learning-based irrigation scheduling using environmental sensor data to predict irrigation requirements and optimize water utilization, achieving 94.74% water utilization efficiency.

TugonLipa: AI-Powered Smart Community Platform (2nd Place in Batangas Province Hackathon) Nov 2025

Technologies: LangChain, RAG, LLM Integration, Automation, Report System

- Developed an AI-powered citizen reporting platform using modern web technologies.
- Integrated AI-based categorization and routing of community reports.
- Designed workflows that automated ticket processing and improved response efficiency.

Autonomous Maze-Solving Robot May 2026

Technologies: ESP32, C++, VL53L0X TOF Sensor, Wall-Following Algorithm, PID

- Implemented autonomous navigation and wall-following algorithms using real-time sensor feedback.
- Integrated TOF sensors and peripherals using I2C communication.
- Developed motor control and motion correction systems using PID feedback control.
- Performed embedded debugging, calibration, and system validation.

EDUCATION

Bachelor of Science in Computer Engineering (Cum Laude) 2022 - 2026

University of Batangas Lipa City | Lipa City, Batangas

SKILLS

Technical Skills

- Programming Languages: Python, JavaScript, TypeScript, PHP, Java, C, C++, SQL
- Embedded Systems: ESP32, Raspberry Pi, Arduino, Firmware Development, Sensor Integration, I2C, SPI, UART, PWM, GPIO, Embedded Debugging
- Web Development: HTML5, CSS3, REST API Integration
- Frameworks & Libraries: Flask, Tailwind CSS, Chart.js, Leaflet.js, Scikit-learn
- Databases: MySQL, PostgreSQL (Supabase), SQLite, MongoDB
- AI, Computer Vision & Automation: YOLO, Roboflow, Dataset Annotation, n8n, LLM Integration, Gemini API
- Networking & IT Support: Hardware & Software Troubleshooting, System Configuration, Technical Documentation
- Tools & Version Control: Git, GitHub, VS Code, Docker, Linux

CERTIFICATIONS & ACHIEVEMENTS

- Cisco CCNAv7: Introduction to Networks
- 2nd Place – Hack the Future: Smart Batangas Province Hackathon Challenge
- Participant – ICpEP.SE Region IV-A CpE Challenge 2025 (Breadboarding)
- Participant – ICpEP.SE Region IV-A CpE Challenge 2024 (C++ Programming)